Disclosed BB84 observations

Bits, basis labels and outcomes contain more than aggregate QBER.

1 IDENTIFY

What do the observations determine?

Gate-path counts
Exact response laws
Rank and row-space tests

Standard and A identify the residual at one setting. B and B+ do not.

Additional observations can restore identification.

2 ESTIMATE

How precisely can it be learned?

Matched scaling ranges
Full operating observations
81-condition precision grid

Full data favor Standard. Scaling usually favors A. Budget and attack level matter.

Compare complete acquisitions at the same signal budget.

3 USE

Does estimated structure improve recovery?

Circuit and pooling controls
New matrix libraries
Fixed-code vs. rate policies

Pooling limits estimation loss. Long-block gains arise mainly through rate selection.

Charge estimation, disclosure and frame-loss costs.

Retain the labels → establish identification → measure precision → test utility